CS 5402: Introduction to Data Mining

SP2026

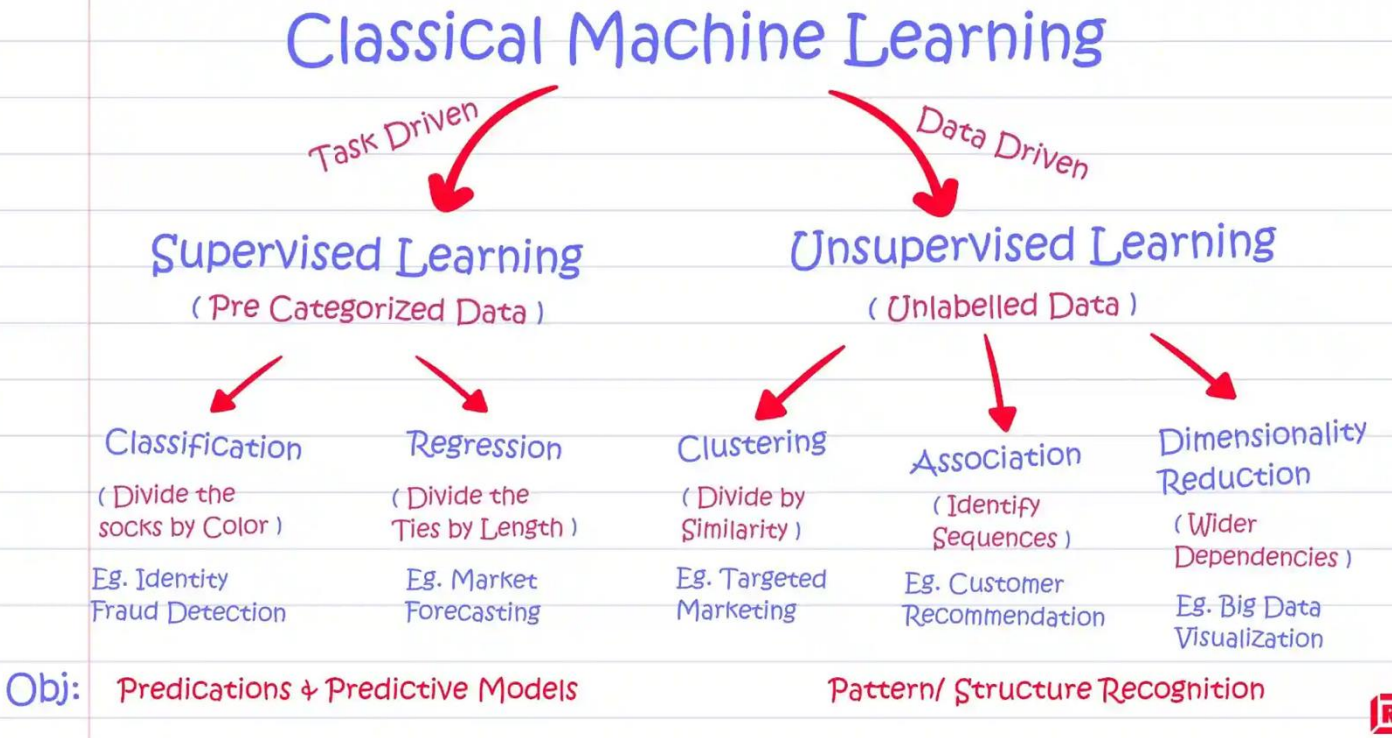
(Lecture - 18)

Self-Supervised Learning (SSL)

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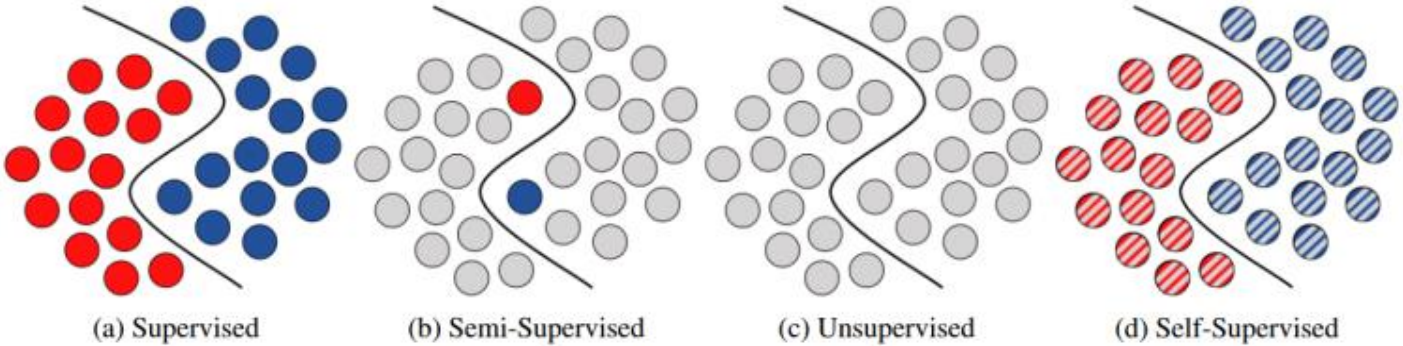
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Overview



https://www.labellerr.com/blog/supervised-vs-unsupervised-learning-whats-the-difference/

Supervised vs. unsupervised vs. self-supervised



A Survey on Semi-, Self- and Unsupervised Learning in Image Classification
https://arxiv.org/pdf/2002.08721.pdf

Problem: Supervised Learning is Expensive! (1)

Assume you want to label **1M** images. How much will it cost?

(1,000,000 images) (Small to medium sized dataset)

× (10 seconds/image) (Fast annotation)

× (1/3600 hours/second) × ($15 / hour) (Low wage paid to annotator)

= $41,667

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

Credit to Justin Johnson, University of Michigan.

Problem: Supervised Learning is Expensive! (2)

Assume you want to label 1B images. How much will it cost?

(1,000,000,000 images) (Small to medium sized dataset)

× (10 seconds/image) (Fast annotation)

× (1/3600 hours/second) × ($15 / hour) (Low wage paid to annotator)

= **$41,666,667**

(Other assumptions: one annotator per image, no benefits / payroll tax / crowdsourcing fee for annotators; not accounting for time to set up tasks for annotators, etc. Real costs could easily be 3x this or more)

Credit to Justin Johnson, University of Michigan.

Medical Image (X-ray)
National average hourly pay for a radiologist is around $218

Problem: Supervised Learning is Not How We Learn.

Babies don't get supervision for everything they see!



The Cake of Learning

How Much Information is the Machine Given during Learning?

Y. LeCun

- **"Pure" Reinforcement Learning (cherry)**
  - The machine predicts a scalar reward given once in a while.
  - **A few bits for some samples**
- **Supervised Learning (icing)**
  - The machine predicts a category or a few numbers for each input
  - Predicting human-supplied data
  - **10→10,000 bits per sample**
- **Self-Supervised Learning (cake génoise)**
  - The machine predicts any part of its input for any observed part.
  - Predicts future frames in videos
  - **Millions of bits per sample**



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The Future is Self-Supervised

Yann LeCun
NYU - Courant Institute & Center for Data Science
Facebook AI Research

Content

- What, why and how?
- Examples of SSL
- Conclusion

What is Self-Supervised Learning (SSL)

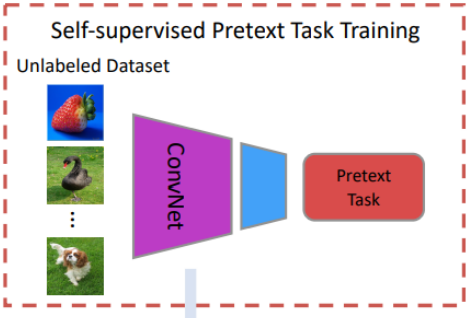
- A subset of unsupervised learning;
  - Learning without using *any human-annotated labels*
- Learn representation from **unlabeled data** through training to solve a pre-text task.
  - Pre-text tasks are pre-designed tasks with pseudo-labels;
  - Pseudo-labels can be generated *without human annotations*.

Why we need it?

- Deep learning show great success on different kinds of tasks (object detection, NLP, medical image analysis, autonomous driving, etc). ***However***, the performance greatly depends on the availability of large amount of training data.
- Collecting and annotating large scale of dataset is an expensive and time-consuming task. On the other hand, the unlabeled data is readily available in abundance.
- Motivation: learn useful representations of the data from a large amount of unlabeled data using self-supervision first, and then fine-tune the representations with few labels for the supervised downstream task.

How SSL works? (Phase-1: Pretraining)

1. Learn useful representation of the data from unlabeled dataset using self-supervised method;

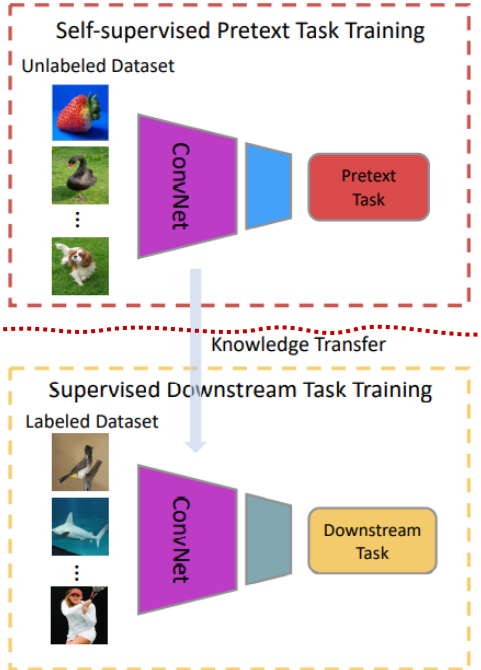


Pre-text task:
- You have a large unlabeled dataset
- Define a task based on the data itself;
- No manual annotation;
- Model is trained like supervised learning;

Image source: [Jing et al., 2019]

How SSL works? (Phase-II: Fine-Tune)

1. Learn useful representation of the data from unlabeled dataset using self-supervised method;



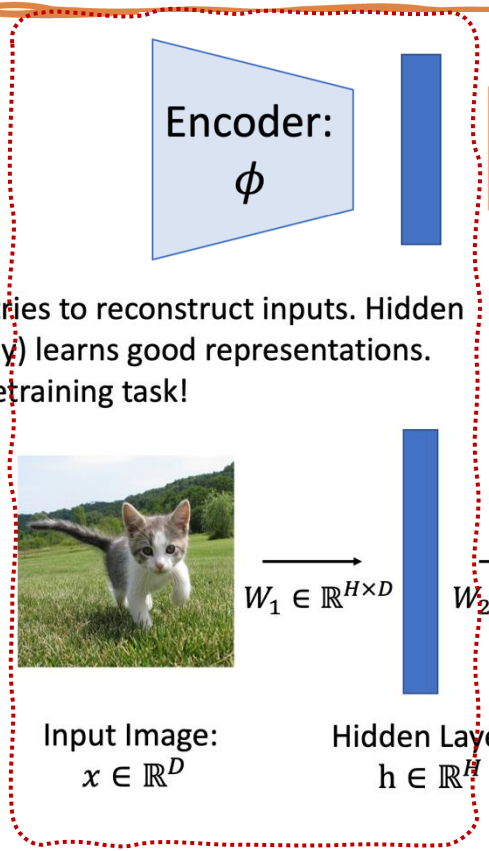
Pre-text task:
- You have a large unlabeled dataset
- Define a task based on the data itself;
- No manual annotation;
- Model is trained like supervised learning;

2. Fine-tune the representations with few labels for the supervised downstream task.

Downstream task:
- The application you care about;
- You do not have large datasets;
- The dataset is labeled

Image source: [Jing et al., 2019]

Recall: Autoencoder



Pretext task: predict rotations (1)

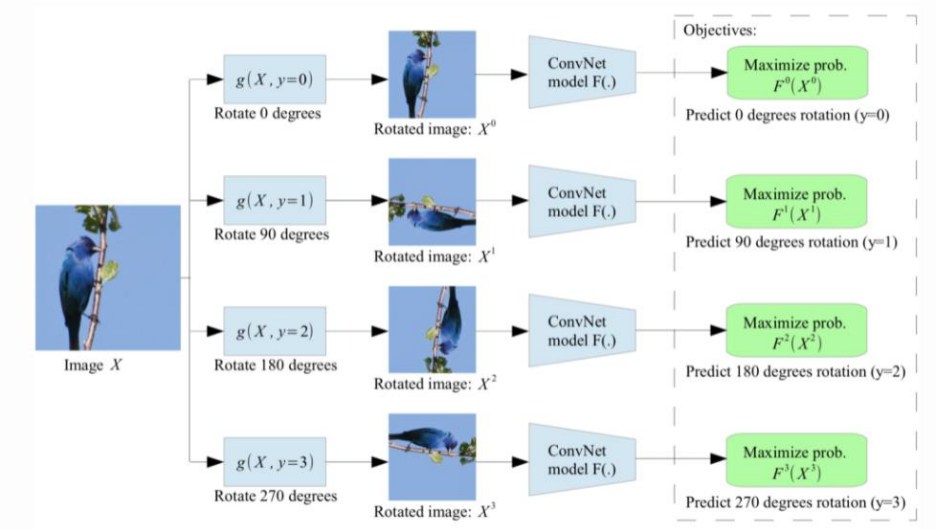
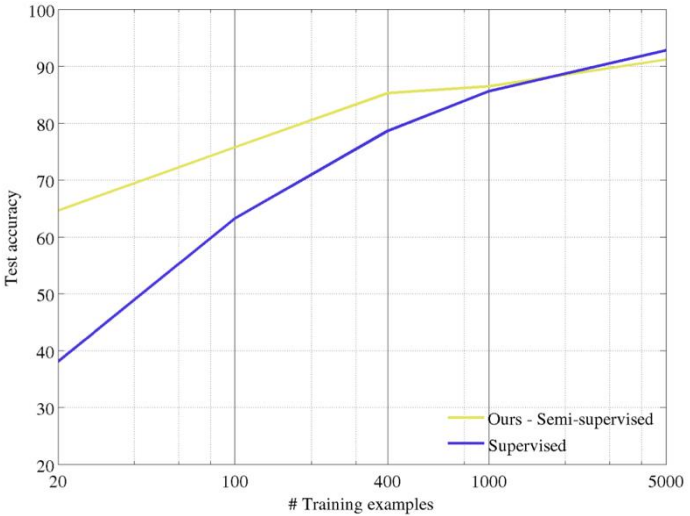


Illustration of self-supervised learning by rotating the entire input images. The model learns to predict which rotation is applied. (Image source: Gidaris et al. 2018)

Hypothesis: a model could recognize the correct rotation of an object only if it has the "visual commonsense" of what the object should look like unperturbed.

Pretext task: predict rotations (2)



(*Image source*: *Gidaris et al. 2018*)

Self-supervised learning on CIFAR10 (entire training set).

Freeze conv1 + conv2
Learn conv3 + linear layers with subset of labeled CIFAR10 data (classification).

Pretext task: predict rotations (3)

Visualize learned visual attentions



(a) **Attention maps of supervised model**

(b) **Attention maps of our self-supervised model**

(*Image source*: *Gidaris et al. 2018*)

Pretext task: predict relative position

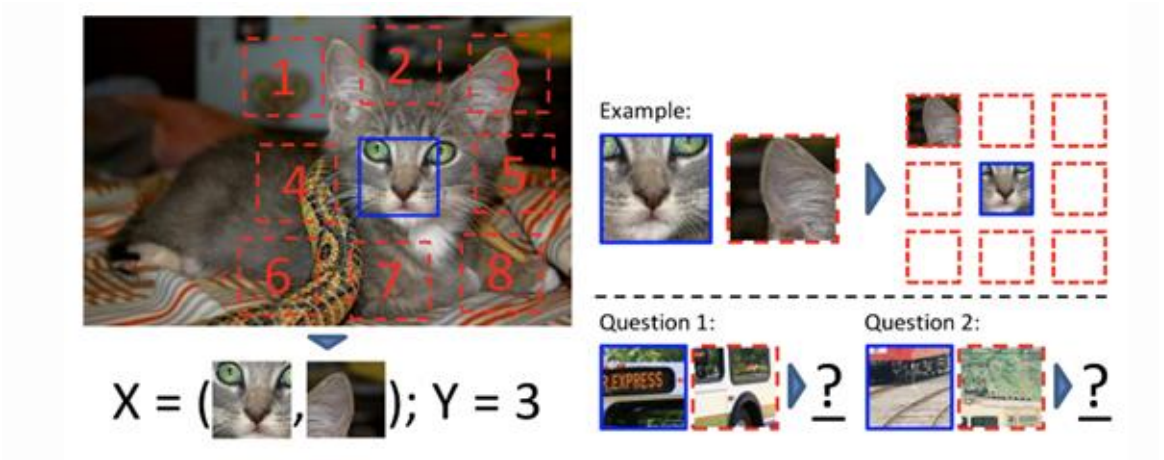
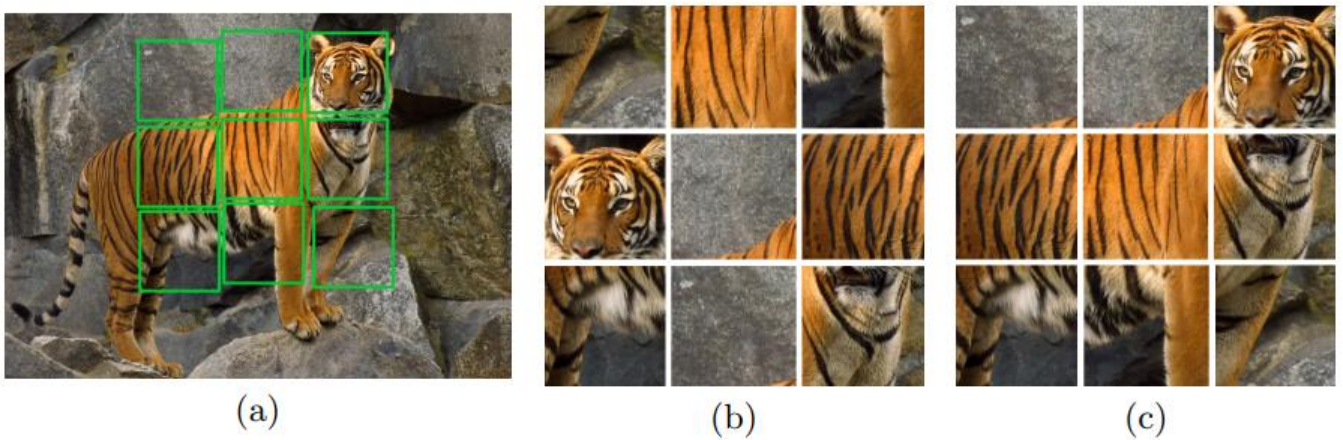


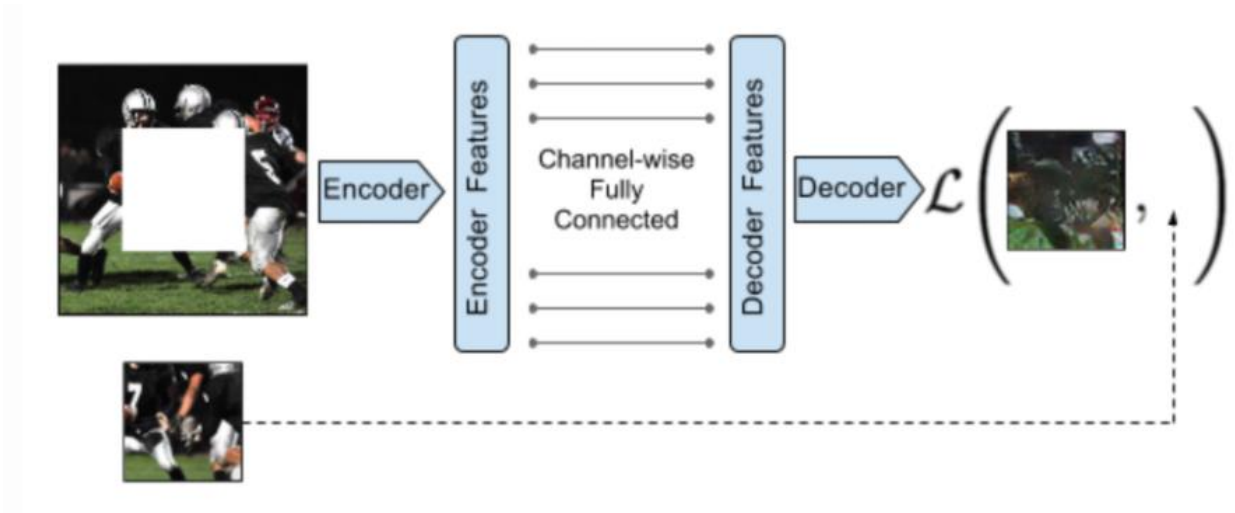
Illustration of self-supervised learning by predicting the relative position of two random patches. Image source: [Doersch et al., 2015]

Pretext task: solving jigsaw puzzle



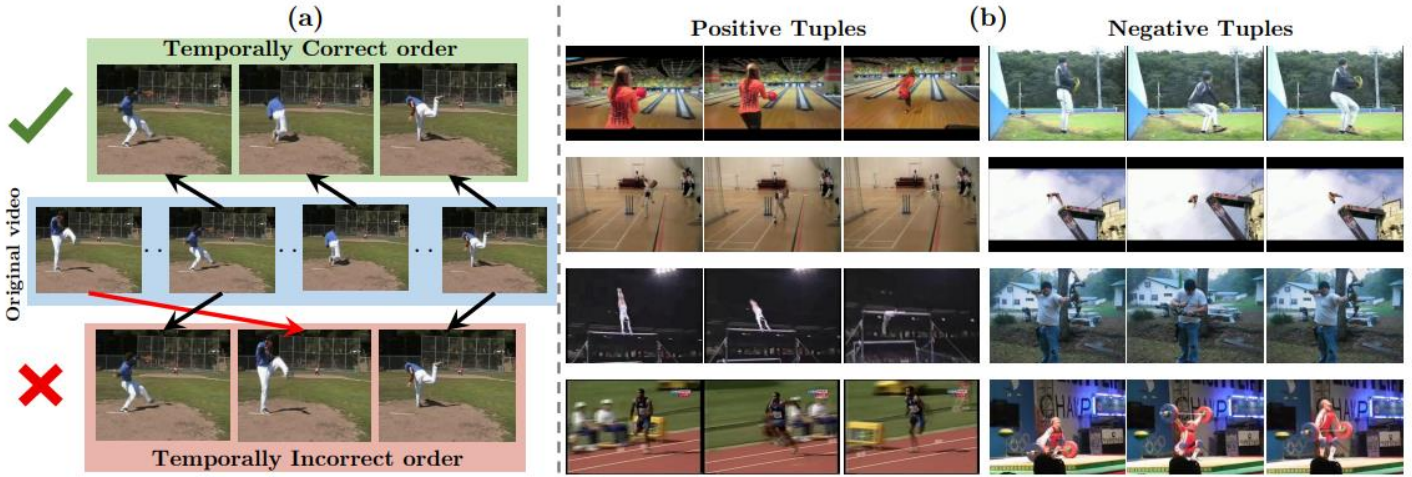
Learning image representations by solving Jigsaw puzzles [*Image source: Noroozi et al 2017*]

Pretext task: predict missing pixels (inpainting)



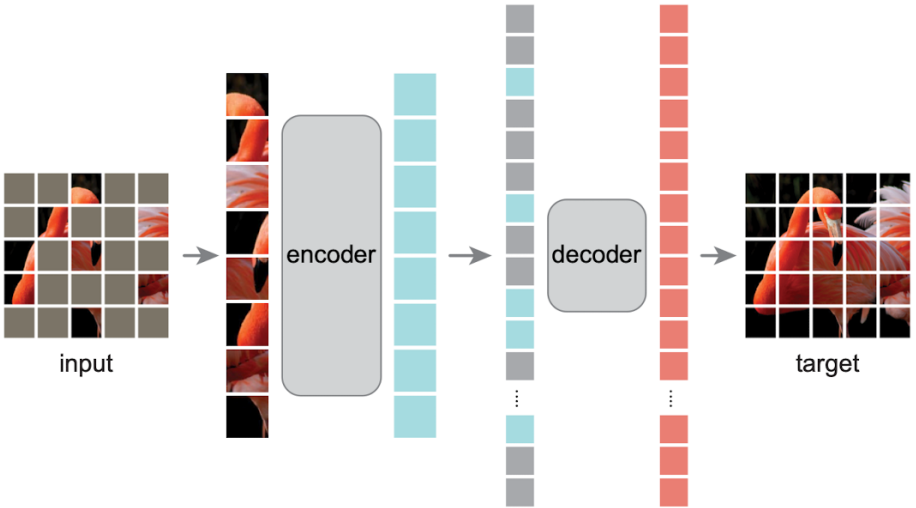
Generative modeling is to reconstruct the original input while learning meaningful latent representation. (*Image source*: Pathak, et al., 2016)

Pretext task: validate order in video frames



Overview of learning representation by validating the order of video frames. (a) the data sample process; (b) the model is a triplet siamese network, where all input frames have shared weights. (Image source: Misra, et al 2016)

Pretext task: Masked Autoencoder

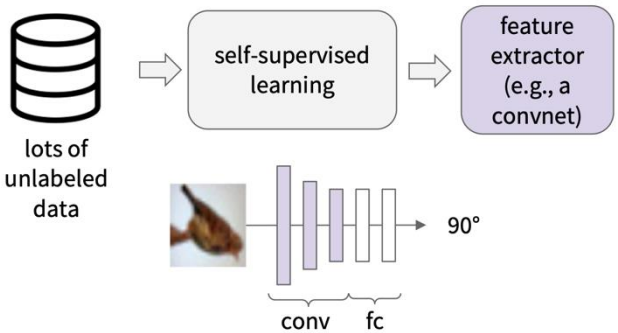


MAE architecture. During pre-training, a large random subset of image patches (e.g., 75%) is masked out. The encoder is applied to the small subset of visible patches. Mask tokens are introduced after the encoder, and the full set of encoded patches and mask tokens is processed by a small decoder that reconstructs the original image in pixels. After pre-training, the decoder is discarded and the encoder is applied to uncorrupted images (full sets of patches) for recognition tasks.. (Haiming, et al 2021)

How to evaluate a self-supervised learning method?

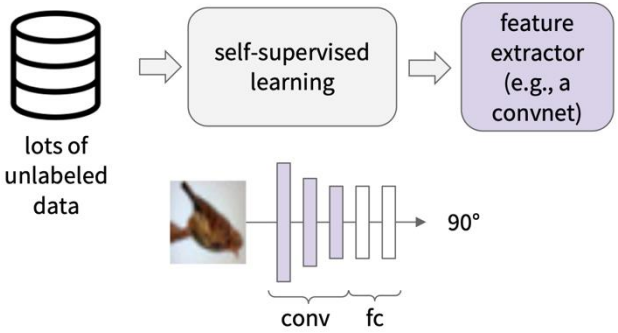
- **Pretext Task Performance**
  Measure how well the model performs on the task it was trained on without labels.
- **Representation Quality**
  Evaluate the quality of the learned representations
  - *Linear Evaluation Protocol:* Train a linear classifier on the learned representations;
  - *Clustering:* Measure clustering performance;
  - *t-SNE:* Visualize the representations to assess their separability.)
- **Robustness and Generalization**
  Test how well the model generalizes to different datasets and is robust to variations.
- **Computational Efficiency**
  Assess the efficiency of the method in terms of training time and resource requirements.
- **Transfer Learning and Downstream Task Performance**
  Assess the utility of the learned representations by transferring them to a downstream supervised task.

Self-supervised learning evaluation: Example (step-1)

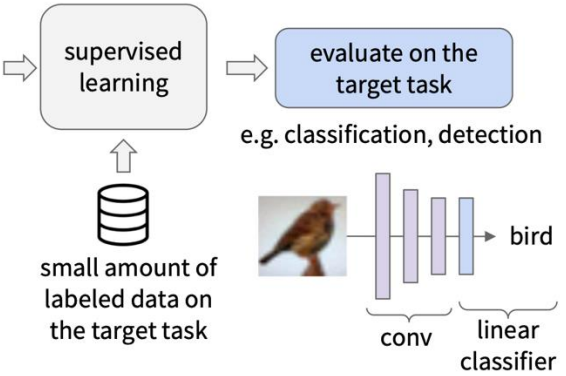


1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations

Self-supervised learning evaluation: Example (step-2)



1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations



2. Attach a shallow network on the feature extractor; train the shallow network on the target task with small amount of labeled data

Summary: pretext tasks from image transformations

- Pretext tasks focus on "visual common sense", e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We don't care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).

Problems:
1. coming up with individual pretext tasks is tedious,
2. the learned representations may not be general.

Self-supervised pretext tasks



Image colorization



Image Super-resolution



Image inpainting



Jigsaw puzzle



Context prediction



Geometric transformation

Learned representations may be tied to a specific pretext task!

Can we come up with a more general pretext task?

From Pretext Tasks → Contrastive Learning

Limitations of Pretext Tasks

- Task-specific design (e.g., rotation prediction, colorization)
- May learn shortcuts instead of meaningful representations
- Limited generalization across tasks

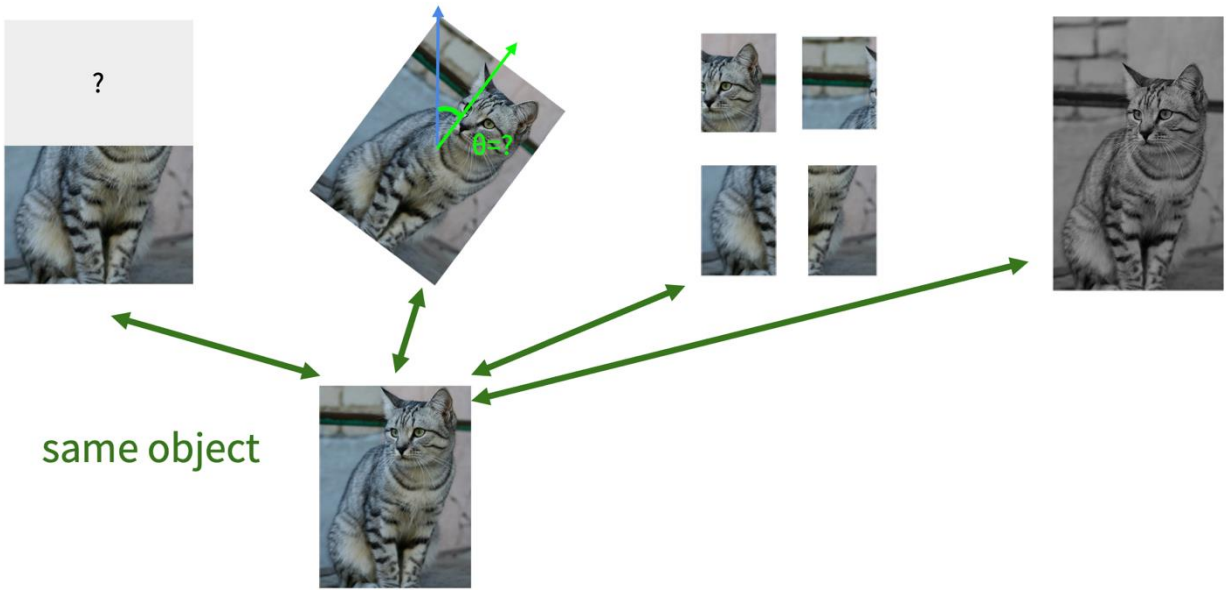
Contrastive Learning

- Learn by comparing similarities and differences
- Pull positive pairs (same instance, different views) closer
- Push negative pairs (different instances) apart

Why It Matters

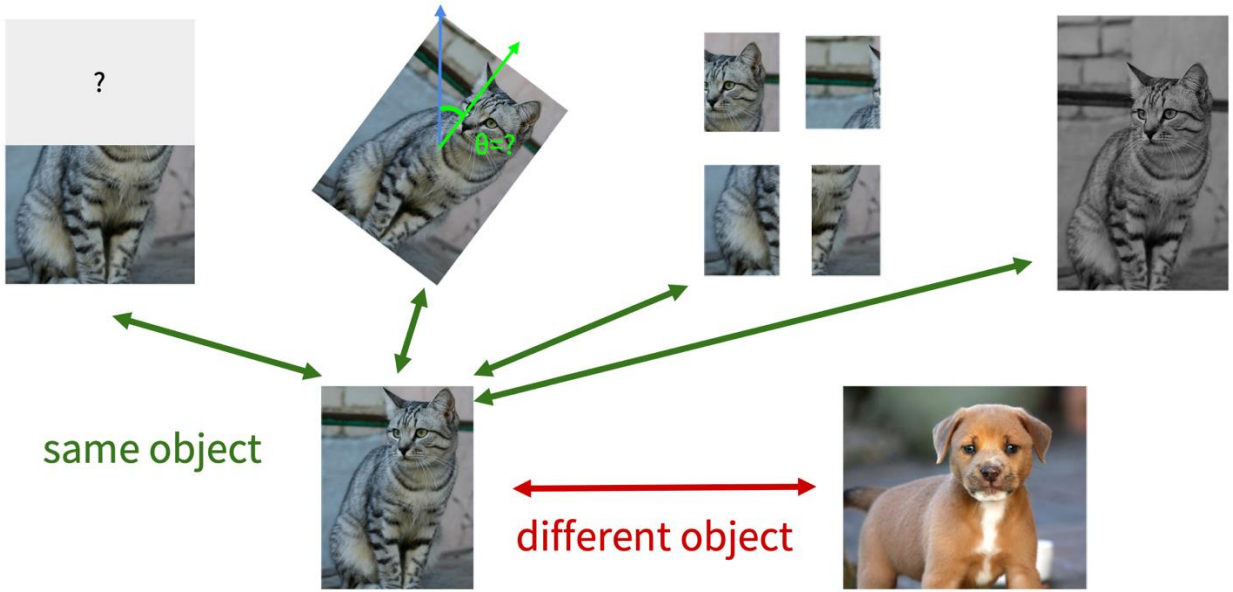
- Learns more robust and general representations
- Reduces reliance on handcrafted pretext tasks
- Foundation for methods like SimCLR, MoCo, BYOL

Contrastive Representation Learning (1)



Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

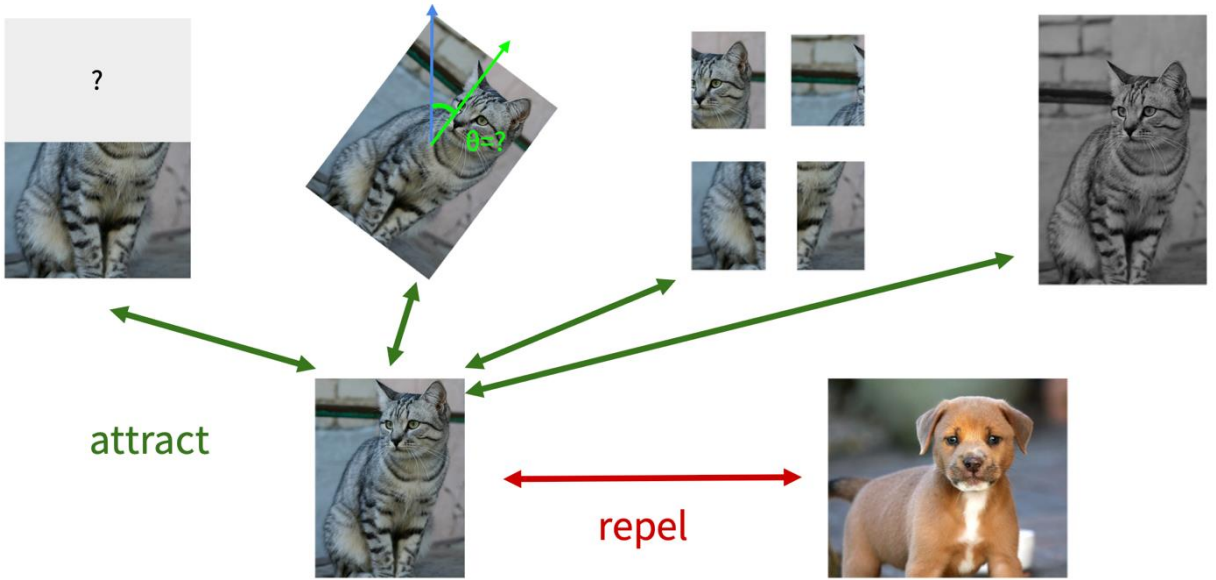
Contrastive Representation Learning (2)



Intuitive: Similar images should have similar features

Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

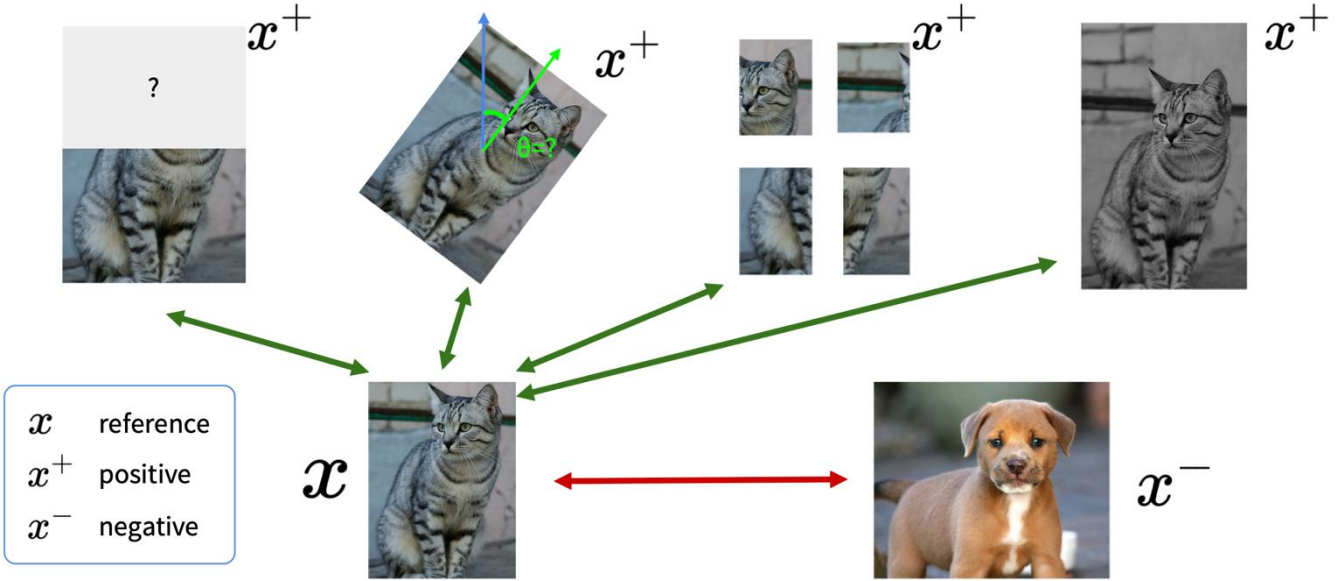
Contrastive Representation Learning (3)



Intuitive: Similar images should have similar features

Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

Contrastive Representation Learning (4)



Intuitive: Similar images should have similar features

Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

Contrastive Representation Learning (5)

Goal:

$$score\big(f(x), f(x^+)\big) \gg score\big(f(x), f(x^-)\big)$$

x: reference sample;
x^+: positive sample.
x^-: negative sample

Given a chosen score function, we aim to learn an encoder function f that yields high score for positive pairs (x, x+) and low scores for negative pairs (x, x-).

Contrastive Representation Learning (6)

A formulation of contrastive learning

Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+))}{\exp(s(f(x), f(x^+)) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-))} \right]$$

Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

Contrastive Representation Learning (7)

A formulation of contrastive learning

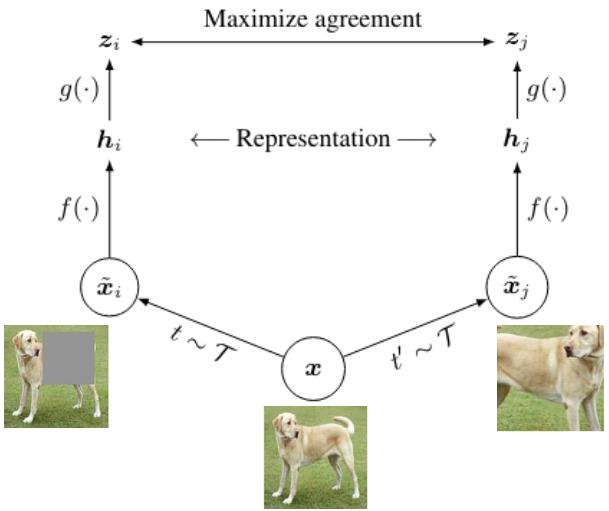
Loss function given 1 positive sample and N - 1 negative samples:

$$L = -\mathbb{E}_X \left[\log \frac{\exp(s(f(x), f(x^+))}{\exp(s(f(x), f(x^+)) + \sum_{j=1}^{N-1} \exp(s(f(x), f(x_j^-))} \right]$$



Credit to Fei-Fei Li and Ehsan Adeli, CS231n, 2024

SimCLR (1)



SimCLR

Cosine similarity as the score function:

$$s(u, v) = \frac{u^T v}{||u||||v||}$$

Use a projection network g(·) to project features to a space where contrastive learning is applied

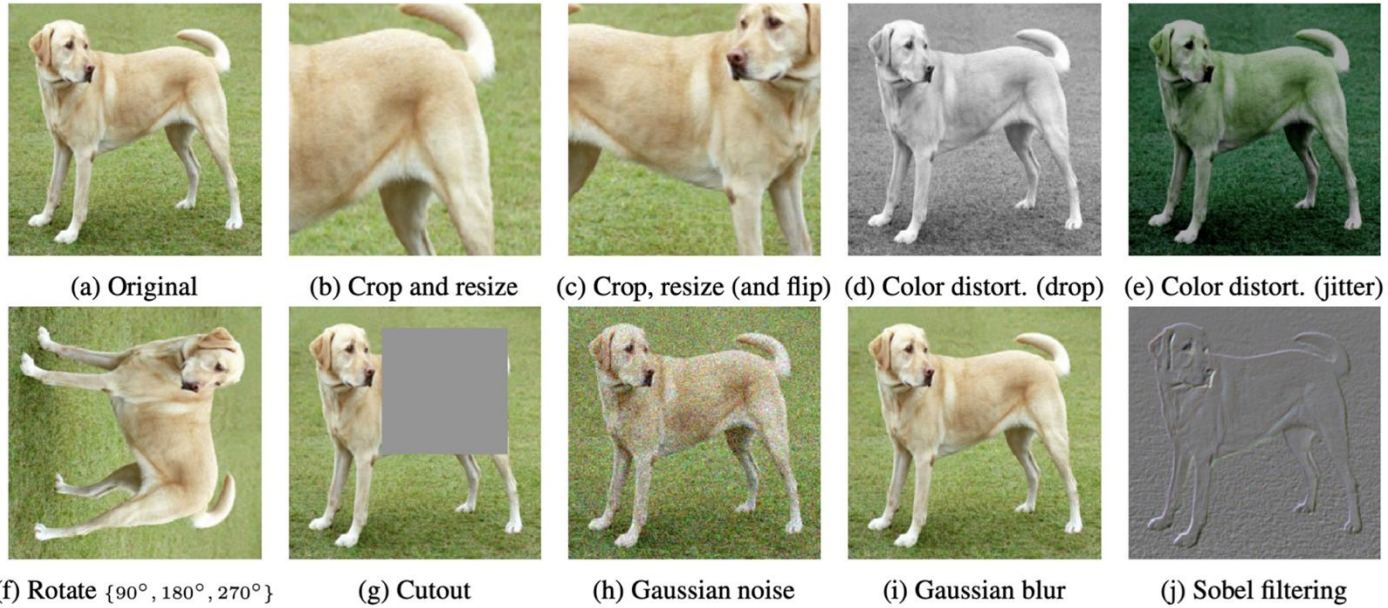
Generate positive samples through data augmentation:

- random cropping, random color distortion, and random blur.

SimCLR: A Simple Framework for Contrastive Learning(Chen, et al. ICML, 2020)

SimCLR (2)

Generating positive samples from data augmentation



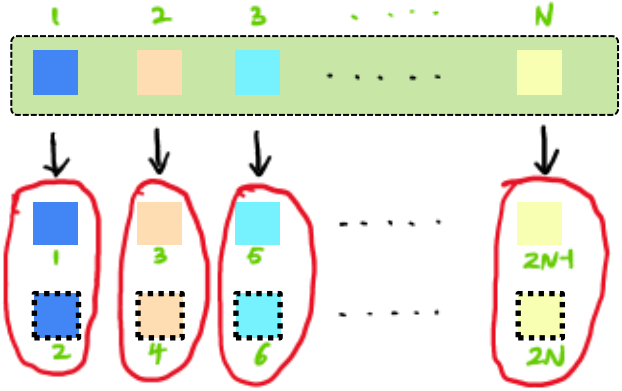
(Chen, et al. ICML, 2020)

SimCLR (3)

Algorithm 1 SimCLR's main learning algorithm.

input: batch size N, constant τ, structure of $f, g, \mathcal{T}$.
for sampled minibatch $\{\boldsymbol{x}_k\}_{k=1}^{N}$ **do**
  for all $k \in \{1, \ldots, N\}$ **do**
    draw two augmentation functions $t \sim \mathcal{T}, t' \sim \mathcal{T}$
    # the first augmentation
    $\tilde{\boldsymbol{x}}_{2k-1} = t(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k-1} = f(\tilde{\boldsymbol{x}}_{2k-1})$ # representation
    $\boldsymbol{z}_{2k-1} = g(\boldsymbol{h}_{2k-1})$ # projection
    # the second augmentation
    $\tilde{\boldsymbol{x}}_{2k} = t'(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k} = f(\tilde{\boldsymbol{x}}_{2k})$ # representation
    $\boldsymbol{z}_{2k} = g(\boldsymbol{h}_{2k})$ # projection
  end for
  for all $i \in \{1, \ldots, 2N\}$ and $j \in \{1, \ldots, 2N\}$ **do**
    $s_{i,j} = \boldsymbol{z}_i^\top \boldsymbol{z}_j / (\|\boldsymbol{z}_i\| \|\boldsymbol{z}_j\|)$ # pairwise similarity
  end for
  define $\ell(i, j)$ **as** $\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
  $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
  update networks f and g to minimize $\mathcal{L}$
end for
return encoder network $f(\cdot)$, and throw away $g(\cdot)$

For a positive pair.



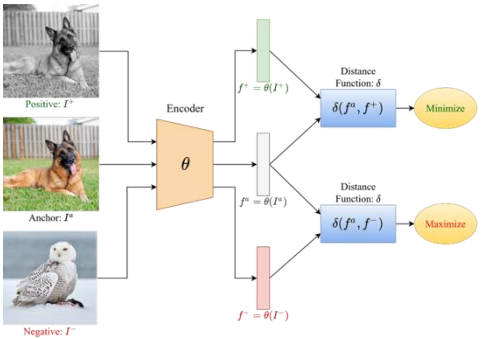
For any sample:
1 positive
-2 Negative

SimCLR (5)

Performance evaluation of fine-tune the pretrained ResNet-50 with few labels on the ImageNet (1M images, 1000 classes)

Method	Architecture	Label fraction 1%	Label fraction 10%
		Top 5	
Supervised baseline	ResNet-50	48.4	80.4
Methods using other label-propagation:			
Pseudo-label	ResNet-50	51.6	82.4
VAT+Entropy Min.	ResNet-50	47.0	83.4
UDA (w. RandAug)	ResNet-50	-	88.5
FixMatch (w. RandAug)	ResNet-50	-	89.1
S4L (Rot+VAT+En. M.)	ResNet-50 (4×)	-	91.2
Methods using representation learning only:			
InstDisc	ResNet-50	39.2	77.4
BigBiGAN	RevNet-50 (4×)	55.2	78.8
PIRL	ResNet-50	57.2	83.8
CPC v2	ResNet-161(*)	77.9	91.2
SimCLR (ours)	ResNet-50	75.5	87.8
SimCLR (ours)	ResNet-50 (2×)	83.0	91.2
SimCLR (ours)	ResNet-50 (4×)	**85.8**	**92.6**

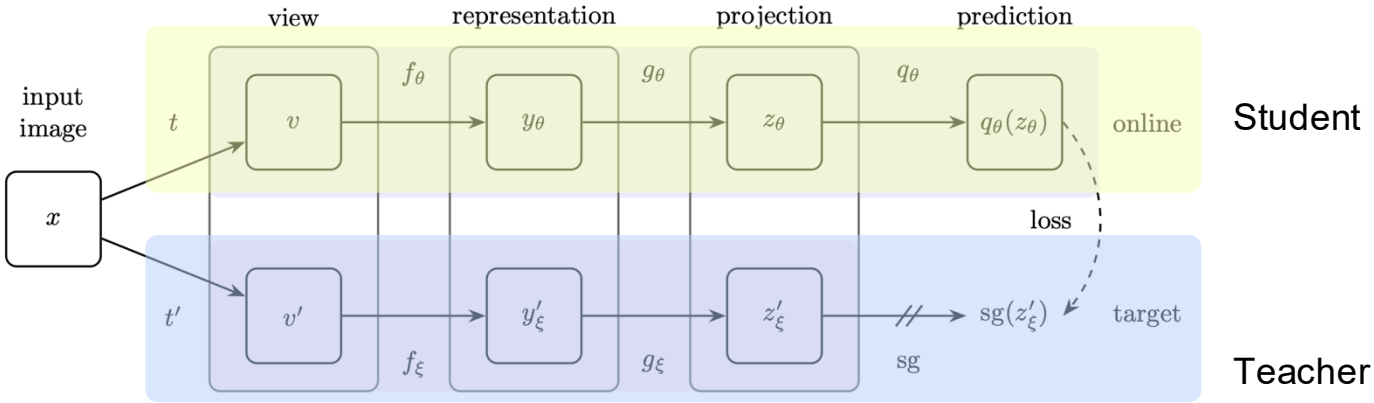
Table 7. ImageNet accuracy of models trained with few labels.



Train feature encoder on ImageNet (entire training set) using SimCLR.

Finetune the encoder with 1% / 10% of labeled data on ImageNet.

BYOL (1)



- Student model learns to predict teacher model's output, using a standard backpropagation to update parameters.

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$$

- Teacher model is updated through Exponential Moving Average:

$$\xi \leftarrow \tau\xi + (1-\tau)\theta \qquad \tau \in [0.99, 0.999]$$

Bootstrap Your Own Latent A New Approach to Self-Supervised Learning [NeurIPS, 2020]

BYOL (2)

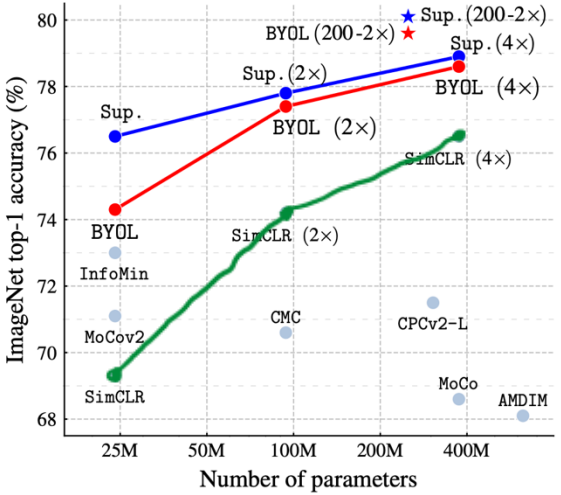
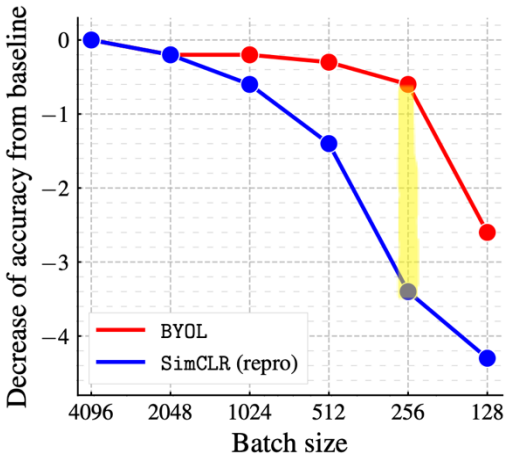


Figure 1: Performance of BYOL on ImageNet (linear evaluation) using ResNet-50 and our best architecture ResNet-200 (2×), compared to other unsupervised and supervised (Sup.) baselines [8].



(a) Impact of batch size

Bootstrap Your Own Latent A New Approach to Self-Supervised Learning [NeurIPS, 2020]

SimCLR vs. BYOL

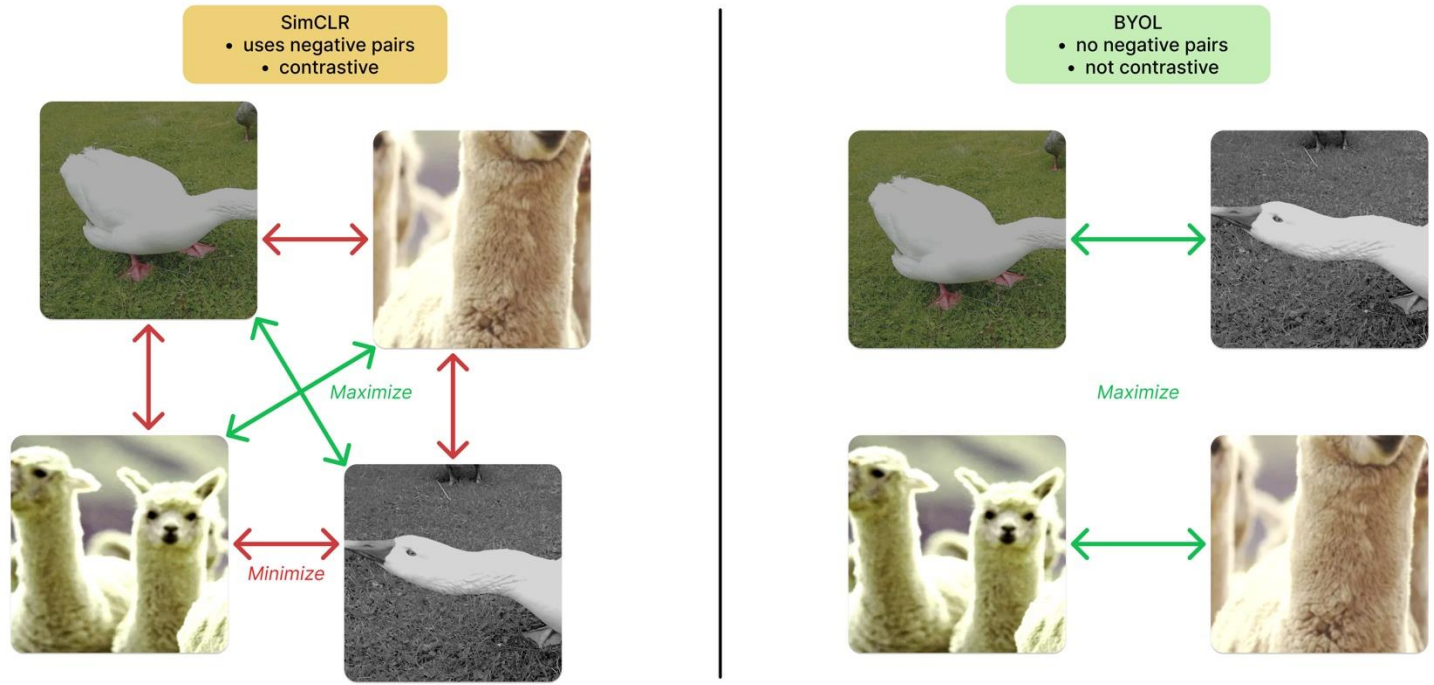
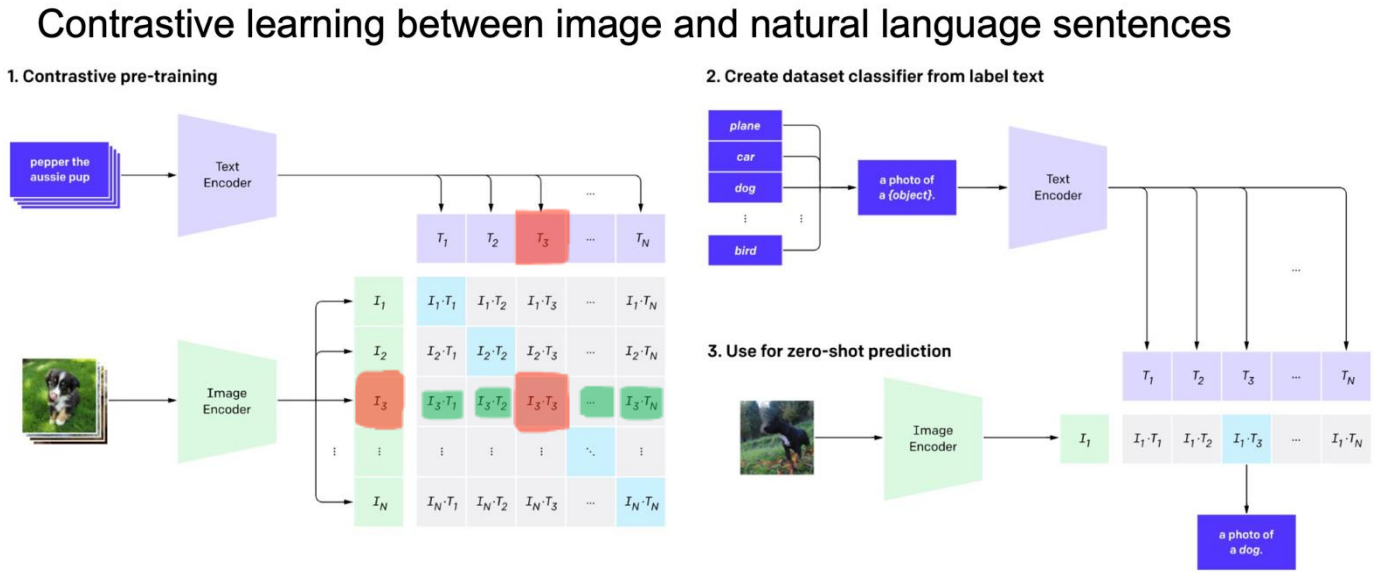


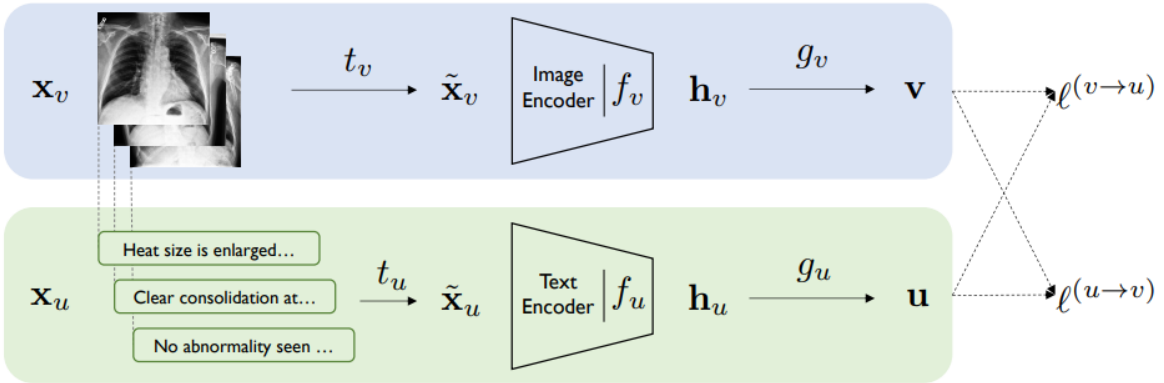
Image credits to: https://mlhonk.substack.com/p/11-byol-bootstrap-your-own-latent

Multimodal Contrastive Learning: Image-Text



CLIP (*Contrastive Language–Image Pre-training*) Radford *et al.*, 2021

Multimodal Contrastive Learning: Medical data



Another example of contrastive learning for medical visual representation . (Image source: Zhang, et al 2020)

Multimodal Contrastive Learning: Time-Series data

Manually construct two view of a single modality

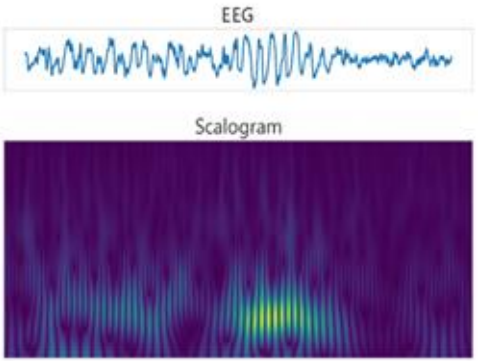


Fig. 1: Illustration of a 30-seconds long electroencephalogram (EEG) signal and a corresponding scalogram extracted with a Morlet wavelet transform.

Raw signals can be converted into spectrogram image!

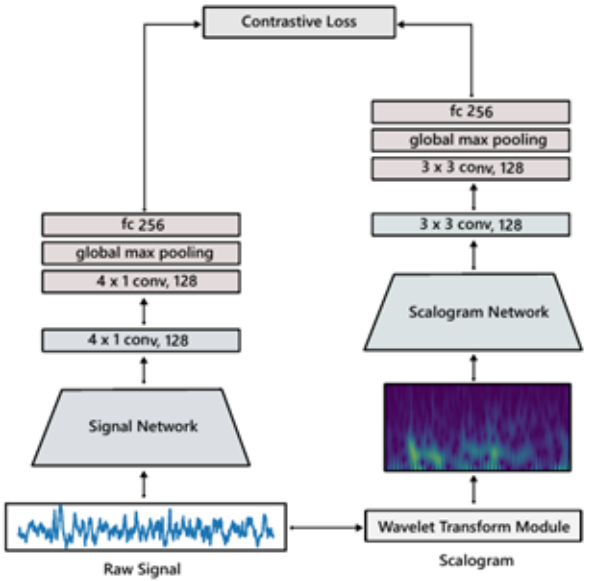
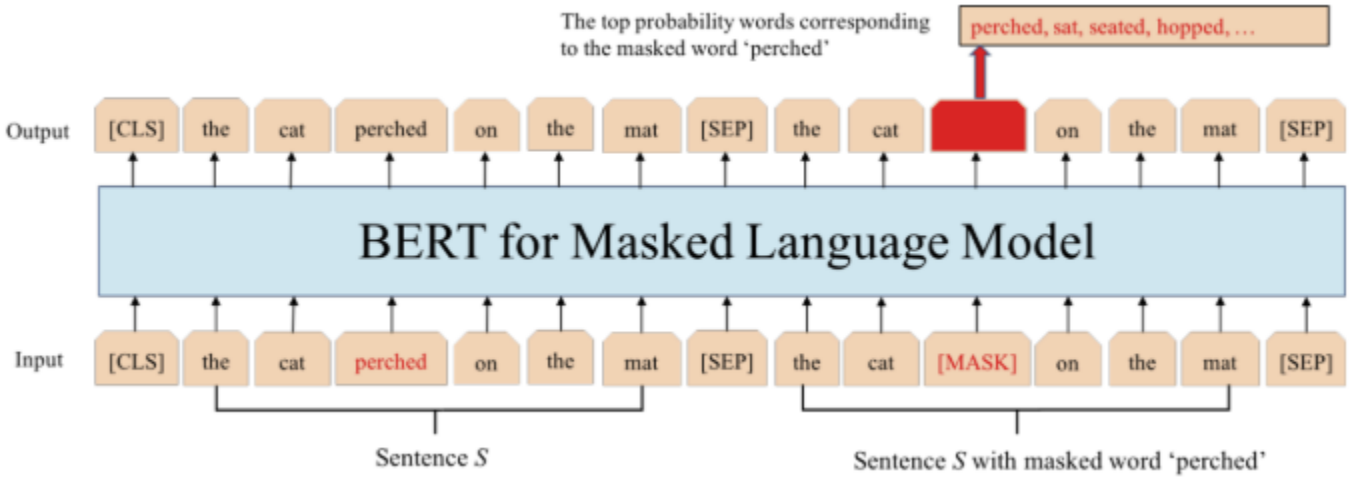


Image source: [Saeed et al. 2020]

More Examples: Text

Pretext task-1: **Masked Language Modeling (MLM)**

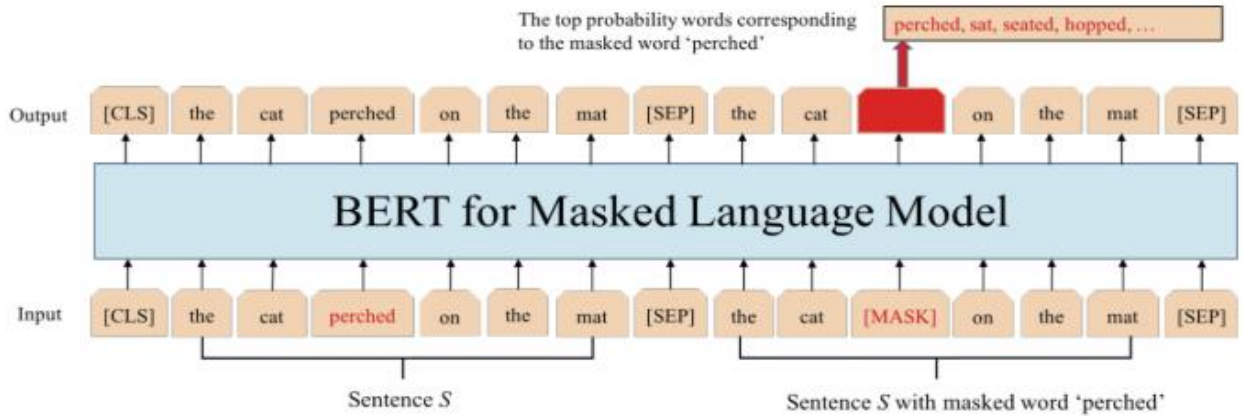


BERT(*Devlin, et al 2019*)

Randomly mask some tokens in a sentence and ask the model to predict them. Typically, 15% of tokens are selected.

More Examples: Text (Cont'd)

Pretext task-2: **Next Sentence Prediction (NSP)**

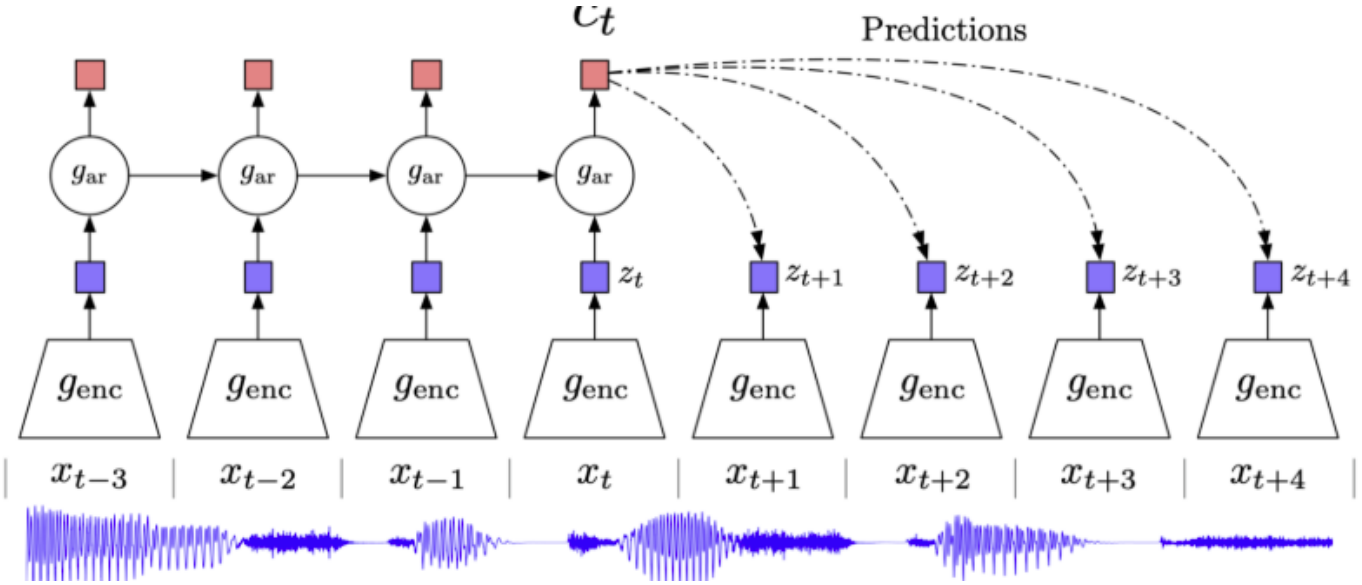


BERT(*Devlin, et al 2019*)

Determine whether two sentences are consecutive in the original text. **Example**
Sentence A: "I went to the store."
Sentence B: "I bought milk." → **IsNext = True**
or
Sentence B: "The sky is blue." → **IsNext = False**

More Examples: Audio

Predicting future from past.



Representation Learning with Contrastive Predictive Coding(Oord, et al 2018)

Conclusion

- Learns from unlabeled data → reduces annotation costs
- Produces transferable, generalizable features
- Pretext tasks enable self-supervision
- High potential for scalable, autonomous learning

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After-Class Questions

- What is the key difference between **supervised**, **unsupervised**, and **self-supervised learning**? Provide an example of each.
- [True or False] Self-supervised learning **does not require any labels**.
- Many SSL methods rely on **predicting part of the data from other parts** (e.g., masked autoencoders). 1. How does this help the model learn useful representations? 2. Can this approach fail in some types of data?
- You have a **time-series dataset** with sensor readings but no labels. Design a self-supervised pretext task to learn useful features. Explain your reasoning.
- Suppose you train an SSL model on a large image dataset. How could you **evaluate the quality of learned representations** without fine-tuning on a downstream task?